Braulio Pantoja Esquina

b_pantojaes@uncg.edu | (984)-325-3165 | www.linkedin.com/in/braulio-pantoja-esquina-8781b61aa/

EDUCATION

University of North Carolina at Greensboro **Greensboro, NC**
Bachelor of Science, Major in Computer Science Expected Graduation May 2026

EXPERIENCE

Artificial Intelligence Researcher **Greensboro, NC**
University of North Carolina at Greensboro September 2025 – Present

- Research and develop multimodal AI models using clinical notes, biomedical knowledge graphs, scientific literature, images, and signal data.
- Apply machine learning techniques to healthcare datasets with emphasis on model reasoning and synthesis across diverse data sources.
- Investigate adaptive, privacy-preserving methods including differential privacy for secure medical AI applications.

Wealth Services Software Engineering Intern **Lake Mary, FL**
Bank of New York Mellon June 2025 – August 2025

- Migrated legacy APIs from MQ to REST, modernizing service architecture for improved scalability and maintainability.
- Developed external-facing APIs using Java and Spring Boot to support real client interactions in Wealth Management systems.
- Created AI-powered agents using python to automate financial tasks and enhance operational efficiency using data from Splunk.
- Built responsive front-end components using Angular, React, HTML, and CSS to deliver seamless user experiences.

Co-Owner **Cary, NC**
Badulake LLC June 2021 – Present

- Evaluated client properties and generated detailed project quotes, supporting successful sales and project planning.
- Managed the company's social media accounts and website, enhancing online visibility and brand image.
- Utilized a self-developed Python lead generation tool to identify potential clients, enhancing outreach efficiency and supporting business growth.

PROJECTS

Greensboro, NC
University of North Carolina at Greensboro January 2026 – Present

- Built and evaluated Python-based disease prediction models using 30 years of historical data from the CDC, WHO, and U.S. Census, achieving strong validation performance (Influenza: AUC 0.9415, Chlamydia: AUC 0.7684).
- Integrated model outputs into the broader web application workflow (JavaScript-based frontend), packaging inference code/artifacts so the team could run reliable end-to-end demos.
- Designed the data ingestion layer to incorporate near–real-time public datasets (NOAA, CDC, U.S. Census) to keep predictions current and improve accuracy.

Lead Generator **Greensboro, NC**
Badulake LLC September 2025 – Present

- Built a Python application that automates finding and qualifying local leads for remodeling, painting, and landscaping services using Google search powered by SerpAPI and web scraping.
- Integrated OpenAI GPT-3.5-Turbo to accurately identify real customer requests, filtering out ads, directories, and irrelevant content.
- Generated over 20 new potential leads on first use, significantly enhancing Badulake LLC's sales pipeline.

Frat App **Greensboro, NC**
Theta Delta Chi International Fraternity – Mu Triton June 2024 – Present

- Developed a Swift-based iOS app to manage fraternity operations, including dues tracking, event scheduling and attendance.
- Integrated calendar APIs and implemented persistent data storage to streamline financial and event workflows.
- Increased on-time dues payments by 31% through automated reminders and real-time tracking features.

LEADERSHIP EXPERIENCE

Theta Delta Chi International Fraternity - Mu Triton Charge **Greensboro, NC**
Executive Board - Librarian May 2025 - Present

- Create and distribute newsletters to alumni featuring chapter news, achievements, and calls for financial and advisory support.
- Lead outreach efforts to foster alumni engagement through mentorship and donations, strengthening chapter-alumni relations.
- Maintain accurate records of chapter history, communications and alumni contact information to support continuity.

Theta Delta Chi International Fraternity – Mu Triton Charge **Greensboro, NC**
Executive Board - Senior Recruitment Chair January 2023 – May 2024

- Increased membership by 40% for Theta Delta Chi Fraternity.
- Delegated tasks and communicated effectively to committee members.
- Report and manage all financial transactions of 16 social and recruitment events.
- Mentoring junior recruitment chair, providing guidance on strategic planning, candidate evaluation, and committee management.

SKILLS

- *Languages – English & Spanish fluency*
- *Technical Skills – Java, Java Script, Assembly, Generative AI, Python, Swift/iOS Development, Jira, Docker, MySQL, Microsoft Office, Flask, Python, PyTorch, NLP, Knowledge Graphs, Differential Privacy, Agile, Splunk, LSTM's*